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No. 26-1/2022-FA/TEC

Date: 14.02.2025

Subject: Minutes of First (1st) meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group (SG)-15 held on 14.02.2025.

The First (1st) meeting of National Working Group (NWG)-15 corresponding to ITU-T Study Group 15 regarding “*Networks, Technologies, and Infrastructures for Transport, Access & home*”, reconstituted for the Study Period 2025-28, was held on 14.02.2025 under the chairmanship of Sh. Rakesh Desai, DDG(FA), TEC through video-conferencing. The list of participants is placed at Annexure A.

2. At the outset, Sh. Jasvir Singh Panesar, Director (FA) TEC & Convener of NWG-15, welcomed all the participants and briefly explained about the agenda and further proceedings of the meeting.

3. Sh. Rakesh Desai, Chair/ NWG-15 & DDG (FA) TEC, greeted all the members and welcomed all participants especially new members from academia, industry and government organisations. He highlighted that the next (4th) meeting of ITU-T SG15 is scheduled at Geneva from **17-28 March 2025**. He pointed out that the last date to upload contributions at ITU-T portal is 4th March 2025 and therefore NWG-15 has to finalise the contributions by 24th March 2025. Members were therefore requested to expedite the finalise their contribution immediately. He further requested the members to attend online correspondence discussion meetings called by the concerned Rapporteurs of SG15 till the third week of February 2025 and update their ongoing discussions. He also emphasized that NWG-15 members and experts have to work upon preparing the contributions which led to the new ITU-T recommendations apart from proposing contributions on revision of the existing recommendations and suggesting some edits in contributions submitted by other member nations. Glimpse of NWG-15 achievements during study period 2022-24 including WTSA-24, present constitution of NWG-15 and ITU-T SG15 structure were briefly presented to the NWG-15 members.

After the introduction, Sh. Rakesh Desai explained the members on potential work-items for contributions that were circulated alongwith meeting notice dated 07.02.2025 and also requested other members to actively participate in the discussions on various contributions. Thereafter, discussions were held on various ongoing contributions presented by the contributors as explained in the succeeding paras.

4. Working document on “Draft to include new Appendix to the ITU-T Rec. L.341 (ex.L.88) on “Management of poles carrying overhead telecommunication lines”. (Q7/15) (Contributors: Sh.Rakesh Desai, Sh. Vishal from M/s Custom Wraps)

Sh.Desai explained the background of this workitem and presented the latest working document shared by the Editor (L.341) wherein revised proposal of NWG-15 was

incorporated (definitions of various terms) in the document. He said that this draft document will be further discussed in the forthcoming meeting of SG15 wherein India has to present again the proposed Appendix to the SG15 experts.

5. Contribution on – Draft new Appendix XII to Recommendation ITU-T G.989.3 (2021) Amd. 1, “Characteristics of Coordinated Multi-Point (CoMP) technology to support IMT-2030/6G and future networks by fronthauling in NG-PON2”. (Q2/15)
(Contributors: Prof. Goutam Das, Preeti Samhita Pati, Sourabh Bhandarkar and Sourav Dutta from IIT Kharagpur, Atri Mukhopadhyay from IIT Bhubaneswar, Sh.Rakesh Desai-TEC)

Mr Atri Mukhopadhyay explained the background of this contribution to the members stating that this proposal was discussed earlier in NWG-15 meetings and SG15 meeting of July 2024 and subsequent correspondence discussions of Q2/15. He presented the draft of new contribution on a new work-item – ‘Draft new Appendix XII to Recommendation ITU-T G.989.3 (2021) Amd. 1, “Characteristics of Coordinated Multi-Point (CoMP) technology to support IMT-2030/6G and future networks by fronthauling in NG-PON2” and also explained the queries of the participants. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by 21st February 2025 for further discussions in the next meeting of NWG-15 scheduled on 24th February 2025 at 4 p.m.

6. Contribution on – Draft new Recommendation L.nis “Practical considerations for network infrastructures sharing” (Q7/15)
(Contributors: Sh.Rakesh Desai & Ms. Mudita Chandra from TEC and Sh. Rajesh Kumar Soni-DOT HQ)

Sh.Desai presented the draft of new contribution on a new work-item – ‘L.nis-Practical considerations for network infrastructures sharing’ and explained the queries of the participants. He stated that India is one of the supporting members of this new work-item. He pointed out that the Q7/15 has invited contribution on ‘National experiences’ in respect of network infrastructures sharing and therefore he requested the members from TSPs / ISPs/IPs to share the information regarding scenarios in network infrastructures sharing so that India can showcase best practices to the other member nations at ITU level. The group experts provided some inputs for improvement of the contribution. Mr Abdul from M/s Bharti Airtel agreed to furnish the information on duct sharing and other network elements. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by 21st February 2025 for further discussions in the next meeting of NWG-15 scheduled on 24th February 2025 at 4 p.m.

7. Contribution on – draft new Supplement of G Suppl.G.65x: Standardization framework for optical fibres for space division multiplexing (SDM). (Q5/15)
(Contributors: Prof. Deepa, Mr. Sobhanbabu and Dr. Chandan Singh Yadav from IIT-Madras, Sh.Arvind from STL, Sh.Rakesh Desai & Ms. Mudita Chandra from TEC)

Dr. Chandan Singh Yadav presented the contribution on behalf of IIT-Madras in respect of draft new Supplement of G Suppl.G.65x and explained the queries of the participants. Sh.Desai highlighted that Multi-Core Fibres (MCF) is being studied in DOT’s ongoing project namely ‘Advance Optical Communication Test Bed Project’ at IIT-Madras and such MCF is being developed by M/s STL. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by 21st February 2025 for further discussions in the next meeting of NWG-15 scheduled on 24th February 2025 at 4 p.m.

8. Contribution on – G.fso : Proposal for update of block diagram of G.fso. (Q6/15)

(Contributors: Sh.Rakesh Desai & Ms. Mudita Chandra from TEC, Dr. A. Paventhan and S Kirubakaran from ERNET, other experts from IIT-Delhi)

Ms. Mudita, ADET (Tx), TEC presented the draft of contribution on ‘Proposal for update of block diagram of G.fso’ and also explained the queries of the participants. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **21st February 2025** for further discussions in the next meeting of NWG-15 scheduled on **24th February 2025 at 4 p.m.**

9. Contribution on – G.fso : Proposal to update the definition for Outage probability for G.fso. (Q6/15)

(Contributors: Sh.Rakesh Desai & Ms. Mudita Chandra from TEC, Dr. Gyan Deep Verma, experts from ERNET and IIT-Delhi)

Ms. Mudita, ADET (Tx), TEC apprised the members about background of this contribution and presented the draft of contribution on ‘Proposal to update the definition for Outage probability for G.fso’. She also explained the queries of the participants. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **21st February 2025** for further discussions in the next meeting of NWG-15 scheduled on **24th February 2025 at 4 p.m.**

10. Contribution on- Proposal for additions and revisions to ITU-T Rec. L.107 “Optical fibre cable construction for sewer duct applications” (Q5/15)

(Contributors: Sh.Rakesh Desai, Sh.Vijay Dixit & Ms. Mudita Chandra from TEC, Sh. Sudipta Bhaumik and Sh. Bikram Nayak from STL)

Ms. Mudita, ADET (Tx), TEC presented the draft of contribution on revisions to ITU-T Rec. L.107 proposing new additional clauses/features on *Electrical Continuity, Material compatibility, Hydrogen etc.* besides other revisions in the working document that was shared by the Editor. She also explained the queries of the participants. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **21st February 2025** for further discussions in the next meeting of NWG-15 scheduled on **24th February 2025 at 4 p.m.**

11. Contribution on- Proposal for additions and revisions to ITU-T Rec. L.108 “Optical fibre cable elements for microduct blowing-installation application” (Q5/15)

(Contributors: Sh.Vijay Dixit & Ms. Mudita Chandra from TEC, Sh. Sudipta Bhaumik and Sh. Bikram Nayak from STL)

Ms. Mudita, ADET (Tx), TEC presented the draft of contribution on revisions to ITU-T Rec. L.108 proposing addition of new clauses/features on *Abrasion test, Biotic damage, bending characteristics of ribbon etc* and exclusion of metallic cable elements in the working document that was shared by the Editor. She also explained the queries of the participants. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **21st February 2025** for further discussions in the next meeting of NWG-15 scheduled on **24th February 2025 at 4 p.m.**

12. Contribution on- Proposal on handling and development of ITU-T

Recommendations for aerial optical fibre cables deployed along electric power lines.

(Q5/15)

(Contributors: Sh.Rakesh Desai & Ms. Mudita Chandra from TEC, Sh. Sudipta Bhaumik and Sh. Bikram Nayak from STL)

Ms. Mudita, ADET (Tx), TEC presented the draft contribution proposing a way forward to deal with development of Recommendations for different types of optical fibre cables deployed along electrical power lines. She further proposed modular approach suggesting five (5) A.1 justifications viz. L.apl: Optical fibre cables for aerial application along electrical power lines; L.apl.1: All Dielectric Self Supporting (ADSS) Optical fibre cables along electrical power lines; L.apl.2: Optical Ground Wire Cable along electrical power lines; L.apl.3: Optical Phase Conductor along electrical power lines and L.apl.4: Optical Attached cables (OPAC) along electrical power lines. She also explained the queries of the participants. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **21st February 2025** for further discussions in the next meeting of NWG-15 scheduled on **24th February 2025 at 4 p.m.**

13. Contribution on- Proposed addition of Appendix II in G.671 version 7.0. (Q6/15)

(Contributors: Ms.Divya Sharma and Sh. Jainendra Bahadur from NSCS)

Sh. Jainendra Bahadur presented the draft of new contribution on addition of Appendix II in G.671 proposing addition of an equipment namely Wavelength detector and tuning device in Appendix and explained the queries of the participants. The group experts observed that the proposed contribution is not proper candidate for the revision of G.671 and the group also provided several inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **21st February 2025** for further discussions in the next meeting of NWG-15 scheduled on **24th February 2025 at 4 p.m.**

14. Contribution on - Revision of Recommendation L.360: “Operations support system requirements for infrastructure and network elements management using Identification data (ID) technology” (Q7/15)

(Contributors: Sh.Rakesh Desai, TEC and S/Sh. Guharajan Shivakumar & Sriram from M/s Aprecomm.)

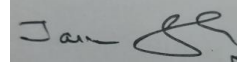
Sh.Desai explained the background of this workitem and apprised the members about past consultations at NWG-15 and SG15 level w.r.to this proposal. Sh.Sriram from M/s Aprecomm presented the updated draft of new contribution on ‘Network Monitoring Expert (NME)’ corresponding to the **Revision of Recommendation L.360** and explained the queries of the participants. The group experts provided some inputs for improvement of the contribution. The contributors were requested to improve upon the draft based on the discussions in the meeting and submit the revised draft contribution by **21st February 2025** for further discussions in the next meeting of NWG-15 scheduled on **24th February 2025 at 4 p.m.**

15. Sh. Rakesh Desai once again requested all the NWG-15 members to obtain TIES account by registering on ITU portal and also register for the forthcoming SG-15 meeting so that members can participate in the discussions during SG15 meeting and can access the desired documents related to contributions at ITU-T level.

16. The key action points emerged from the meeting are as under: -

- i. Members / experts are requested to submit contributions as per the timeline as discussed above.

- ii. Next (2nd) meeting of NWG-15 will be held at 4 p.m. on Monday, 24th February 2025. Meeting link will remain same i.e <https://cdotmeet.cdot.in/vmeet/rooms/1bb-a8c-ggt-ucn/join> .
17. The meeting ended with the vote of thanks to the chair.



(Jasvir Singh Panesar)
Convener, NWG-15 & Director (FA), TEC

To,

All the participants / Members of the NWG-15.
Copy to: Sr.DDG & Head, TEC for kind information.

Annexure- 'A'**List of Participants**

1. Sh. Rakesh Desai, DDG (FA), TEC
2. Sh. J.B Chavan, DDG (WR), DoT
3. Sh. S K Bhalla, DDG, NE LSA
4. Sh. Mukesh Kumar, DDG Radio, TEC
5. Sh. Rajesh Kumar Soni, DDG (CS),DoT
6. Sh. Jasvir Singh Panesar, Dir (FA), TEC
7. Sh. Varughese Mathew, BSNL
8. Sh. Gyan Deep Verma, M/s Kaust
9. Sh. Sudipto Das, M/s Phoenix Contact India
10. Sh. Vivek Dixit, M/s Ericsson India Pvt. Ltd.
11. Dr. Lenin Raja, AA Electro Magnetic Test Lab
12. Sh. Sudipta Bhaumik, M/s Sterlite Technologies Ltd.
13. Sh. Chandan Singh Yadav, IIT Madars
14. Sh. Sobhanbabu, IIT Madars
15. Sh. Rahul Dhar, M/s Raultech
16. Sh. Jainendra, NSCS
17. Sh. Atri Mukhopadhyaya , IIT Bhubansewar
18. Ms. Mudita Chandra, ADET (Tx), TEC
19. Ms. Yashita Mudgal, RA(FA), TEC
20. Sh. Digamber Yadav, C-DoT
21. Sh. Suresh Kumar, M/s Susan Future Technologies Pvt. Ltd.
22. Sh. Sourabh Bhandarkar, IIT Kharagpur
23. Sh. Abdul Mateen Ashfaq , Bharti Airtel
24. Prof.Deepa Venkitesh, IIT Madras
25. Sh. Srirama Murty Nistala, Aprecomm Pvt Ltd
26. Sh. Kirubakaran, ERNET India
27. Sh. Abhijit, IIIT Delhi
28. Ms.Anjali Gupta, IIT Delhi
29. Sh. Bikram Kesharee Nayak M/s Sterlite Technologies Pvt. Ltd.
30. Preeti ,IIT KGP
31. Sh. Roshan Kumar, M/s West Coast Optilinks
32. Sh. Goutam Das, IIT Kharagpur
33. Dr. A. Paventhan, ERNET India
34. Deo Pratap, AD(FA), TEC
